

Lab 7

Mixed finite elements (Darcy equations). deal.II step-20.

Advanced Topic in Scientific Computing - SISSA, UniTS, 2026

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(Based on notes by P. C. Africa)

Assignment

Read the documentation of `step-20`.

1. Modify `step-20` so to use matrices and vectors from the `TrilinosWrappers` namespace.
2. In the `solve()` method, try different solvers and preconditioners in order to detect the combination minimizing the number of linear solver iterations. Focus in particular on Algebraic Multi-Grid (AMG) preconditioners.